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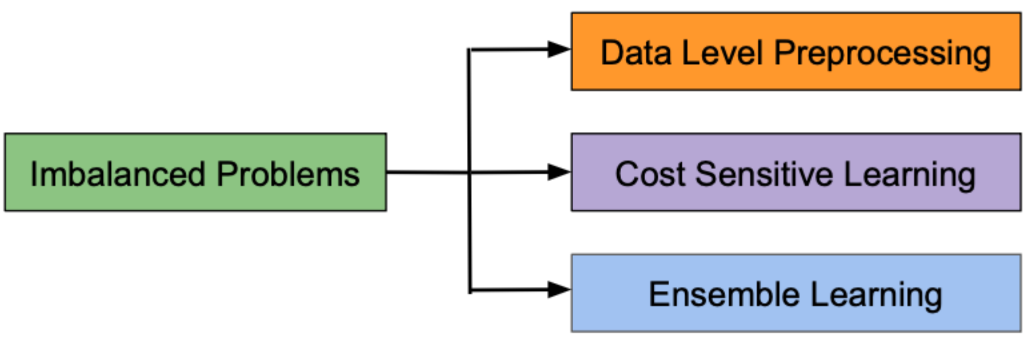
Imbalanced Problems

In Machine Learning and Deep Learning is a very common problem, the Ïmbalanced Problem". To adress this problem we have mainly three strategies.

i. Data Level Preprocessing: This is Undersampling, Oversampling, among others.

ii. Cost Sensitive Learning: We are going to adress this strategy in this document.

iii. Ensemble Learning: Here we have the the well known models created using ensemble learnings.



Cost Sensitive Learning:

For normal training for classification problems we have:

$$\boxed{Cost = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \cdot ln(\hat{y_i}) + (1 - y_i) \cdot ln(1 - \hat{y_i})\right]}$$

And we want to minize this loss functions, but in <u>Cost Sensitive Learning</u> we want to minize:

$$\boxed{Cost = -\frac{1}{n}\sum_{i=1}^{n}\left[w_1 \cdot y_i \cdot ln(\hat{y_i}) + w_0 \cdot (1 - y_i) \cdot ln(1 - \hat{y_i})\right]}$$

Where:

$$\boxed{w_j = \frac{n}{2 \cdot n_j} = \left(\left(\frac{1}{2}\right) \cdot \left(\frac{n_j}{n}\right)\right)^{-1}}$$
